Solutions

Quiz 9

1. Consider a driven damped oscillator with the following equation of motion:

$$Dx = f_0 \cos(\omega t) \tag{1}$$

where

$$D = \frac{d^2}{dt^2} + 2\beta \frac{d}{dt} + \omega_0^2 \,. \tag{2}$$

Solving the equation of motion yields, for this oscillator:

$$x(t) = \frac{f_0}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + 4\beta^2\omega^2}} \cos\left(\omega t - \tan^{-1}\left[\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right]\right) \tag{3}$$

(a) Find the solution $x_n(t)$ to the following equation of motion:

$$Dx_n = f_n \cos(n\omega t)\,, \tag{4}$$

where D is given in Eq. (2).

$$x_n(t) = \frac{f_n}{(\omega_0 - n^2\omega^2)^2 + 4\beta^2 n^2 \omega^2} \cos\left(n\omega t - \tan^{-1}\left[\frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right]\right)$$

(b) Find the solution $x(t)$ to the following equation of motion:

$$Dx = \sum_{n=0}^{\infty} a_n \cos(n\omega t)\,, \tag{5}$$

where D is given in Eq. (2).

$$x(t) = \sum_{n=0}^{\infty} \frac{f_n}{(\omega_0 - n^2\omega^2)^2 + 4\beta^2 n^2 \omega^2} \cos\left(n\omega t - \tan^{-1}\left[\frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right]\right)$$

2. Consider the length of a general curve

$$L = \int_A^B ds\,,$$

where $ds = \sqrt{d\vec{r} \cdot d\vec{r}}$. Let this curve constrained to a flat 2D plane. Only consider curves that can be written as $y(x)$.

(a) Rewrite ds in terms of dx and dy

$d\vec{r} = dx\,\hat{x} + dy\,\hat{y}$

$d\vec{r} \cdot d\vec{r} = dx^2 + dy^2$

$$\boxed{ds = \sqrt{dx^2 + dy^2}}$$

(b) Fill in the blank

$$L = \int_{x_A}^{x_B} \underline{\sqrt{1 + [y'(x)]^2}}\; dx$$

Leave your answer in terms of $y(x)$, $y'(x)$, x, and dx.

$$= \sqrt{(dx^2)\left(1 + \frac{dy^2}{dx^2}\right)}$$

$$= dx\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= dx\sqrt{1 + [y'(x)]^2}$$